Quantum Information

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Abstract

Extremely rough notes on quantum information. Maybe these will turn into something coherent, maybe not!

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1 POVM Measurements

1.1 Pure States and Projective Measurements

As typically introduced, the quantum mechanics of pure states $|\psi\rangle$ living in a Hilbert space $\mathcal{H}$ connects to experiments via projective measurements. Formally, a projective measurement is described by an indexed collection of projectors

$$\{\Pi_x : \mathcal{H} \to \mathcal{H} \mid x \in S\}, \quad S \subset \mathbb{R} \tag{1}$$

which add up[1] to the identity

$$\sum_{x \in S} \Pi_x = \mathbf{1} \tag{2}$$

so that they equivalently specify an internal direct sum decomposition

$$\mathcal{H} = \bigoplus_{x \in S} \mathcal{H}_s, \quad \mathcal{H}_s \equiv \Pi_x(\mathcal{H}). \tag{3}$$

A projective measurement then randomly projects any prepared pure state $|\psi\rangle \in \mathcal{H}$

$$|\psi\rangle \xmapsto{\text{proj}} \frac{\Pi_x |\psi\rangle}{\|\Pi_x |\psi\rangle\|} = \frac{\Pi_x |\psi\rangle}{\sqrt{\langle\psi| \Pi_x |\psi\rangle}} \tag{4}$$

and the corresponding $x \in S$ is called the measurement outcome. Assuming WLOG that $|\psi\rangle$ is properly normalized $\langle\psi|\psi\rangle = 1$, the probability of any particular outcome is given by the Born rule

$$\text{Prob}_x = \|\Pi_x |\psi\rangle\|^2 = \langle\psi| \Pi_x |\psi\rangle \,. \tag{5}$$

If $S \subset \mathbb{R}$ is anything more analytically interesting than a finite set, this object should be interpreted with care as a probability measure. For the present discussion, it suffices to imagine that S is set.

[1] Here "addition" is meant generously to include integration.

1.2 Entangled States and POVMs

The above is all well and good, but the following simple and natural thought experiment suggests a more general kind of process which does not quite take the form of a projective measurement, but which intuitively warrants being called a "measurement" of our system. Suppose we are interested in the physics of some Hilbert space $\mathcal{H}_A$, but that there exists a second space $\mathcal{H}_B$ and that we have the experimental power to perform general joint measurements. A general joint state

$$|\Psi\rangle_{AB} \in \mathcal{H}_A \otimes \mathcal{H}_B \tag{6}$$

is entangled, but for our purposes it will suffice to consider a pure state

$$|\Psi\rangle_{AB} = |\psi\rangle \otimes |\phi_{\text{aux}}\rangle \tag{7}$$

where we think of the second factor as an auxiliary state. Now suppose we perform a projective measurement on the joint state, described as above by a set of projectors

$$\{\Pi_{AB,x} \mid x \in S \subset \mathbb{R}\}. \tag{8}$$

The Born rule applied to the joint state gives the probability of outcome $x \in S$ as

$$\text{Prob}_x = \left(\langle\psi| \otimes \langle\phi_{\text{aux}}|\right) \Pi_{AB,x} \left(|\psi\rangle \otimes |\phi_{\text{aux}}\rangle\right). \tag{9}$$

If we think of this as a function of $|\psi\rangle$ at fixed $|\Phi_{\text{aux}}\rangle$ and $\Pi_{AB,x}$, we can naturally rewrite the result as the diagonal matrix element of a quadratic form Q_x on $\mathcal{H}$ with matrix elements

$$Q_x : \mathcal{X}_A \times \mathcal{H}_A \to \mathbb{C}, \quad \langle\phi| Q_x |\chi\rangle \equiv \left(\langle\phi| \otimes \langle\phi_{\text{aux}}|\right) \Pi_{AB,x} \left(|\chi\rangle \otimes |\phi_{\text{aux}}\rangle\right). \tag{10}$$

The RHS is antilinear in the first argument and linear in the second, so this defines a bona fide quadratic form, and we can identify

$$\text{Prob}_x = \left(\langle\psi| \otimes \langle\phi_{\text{aux}}|\right) \Pi_{AB,x} \left(|\psi\rangle \otimes |\phi_{\text{aux}}\rangle\right) = \langle\psi| Q_x |\psi\rangle. \tag{11}$$

Recall that the polarization identity on quadratic forms

$$Q(u,v) = \frac{Q(u+v,u+v) - Q(u,u) - Q(v,v)}{2} \tag{12}$$

tells us a quadratic form's entire action is fixed by its diagonal values, so knowing the value of $\langle\psi| Q_x |\psi\rangle$ for all $|\psi\rangle \in \mathcal{H}_A$ (or more weakly by linearity, just its values on unit-normalized such states) fixes Q_x uniquely.

The defining properties of $\Pi_{AX,b}$ as a projective measurement, plus judicious use of the polarization identity, imply that each Q_x is a Hermitian positive semi-definite operator[2]

$$Q_x = Q_x^\dagger, \quad Q_x \geq 0 \tag{13}$$

and that the collection taken together is a decomposition of the identity

$$\sum_{x \in S} Q_x = \mathbf{1}_A. \tag{14}$$

These two properties define a *positive operator-valued measurement* (POVM). The reader can convince themselves that any such collection $\{Q_x\}_{x\in S}$ with these two properties can be cooked up as we did here, by a suitable choice of auxiliary state $|\phi_{\text{aux}}\rangle \in \mathcal{H}_B$ and joint projective measurement $\{\Pi_{AB,x}\}_{x\in S}$.

[2]Here we are slightly redundant, as positive semi-definiteness is only defined for Hermitian operators to begin with.

2 Pure State Estimation, Symmetric Subspace

In the words of Michael Walter, we new address the

Goal *(Pure State Estimation): Find a POVM* $\{Q_{\hat{\phi}}\}_{\hat{\phi}\in\Omega}$ *on* $(\mathbb{C}^d)^{\otimes n}$*), with possible outcomes in the set*

$$\Omega = \left\{\rho_{\hat{\phi}} = \left|\hat{\phi}\right\rangle\left\langle\hat{\phi}\right|\right\} \tag{15}$$

of pure states on $\mathbb{C}^d$*, such that when we measure* $\phi^{\otimes n} = |\phi\rangle^{\otimes n}\langle\phi|^{\otimes n}$*, we obtain an outcome* $\hat{\phi}$ *that is "close" to* ϕ*.*

As a small clarification: when we talk about outcomes, we are not saying that the POVM projects down to pure states on $\mathbb{C}^d$. Indeed, in general a POVM does not admit an interpretation as a projector with a pure state measurement outcome. We mean that we have some collection of pure states in $\mathbb{C}^d$; that we are using this collection as our indexing set for the POVM; and that when our measurement returns the outcome associated with $\left|\hat{\phi}\right\rangle$, we interpret this as our best guess for the state whose n-fold product is our initial pre-measurement state. A well-designed such POVM at variable n should have the property that, in the limit of large n, this procedure should yield guesses $\left|\hat{\phi}\right\rangle$ which are extremely similar to $|\phi\rangle$ extremely often. This is plausible essentially because, given two unit-normalized states $|\psi\rangle, |\psi\rangle \in \mathbb{C}^d$, we have

$$|\langle\psi|\xi\rangle| < 1 \implies \left|\left(\langle\psi|^{\otimes n}\right)\left(|\xi\rangle^{\otimes n}\right)\right| = |\langle\psi|\xi\rangle|^n \xrightarrow{n\to\infty} 0. \tag{16}$$

So if two states are at all distinguishable, then their corresponding n-fold copies are very distinguishable for large enough n. Of course the order of predicates is important. This is a pointwise limit (for any two fixed distinct states, distinguishability goes to 1 as n goes to ∞), not a uniform one (to the contrary, given any n no matter how large, we could adversarially produce distinct states whose $n$fold products are arbitrarily hard to distinguish).

2.1 Continuous POVMs and the Uniform Measure

In order to get a typical guess $\left|\hat{\phi}\right\rangle$ which is arbitrarily close to the true state $|\phi\rangle$ at large n no matter what the latter happens to be, our indexing set $\Omega \subset \mathbb{C}^d$ must at least contain states arbitrarily close to any possible true state. Rather than taking Ω itself to depend on n, we will take Ω from the state to be uncountably infinite, covering all distinct states in $\mathbb{C}^d$. As two states are distinct if they correspond to different rays, this means our indexing set is $\Omega \simeq \mathbb{CP}^{d-1}$.

Now the correct probabilistic interpretation of our outcomes must be more explicitly measure-theoretic. A continuous POVM is specified by an indexed collection of operators on $\mathcal{H}$

$$\{Q_x : \mathcal{H} \to \mathcal{H}\}_{x\in\Omega} \tag{17}$$

plus the auxiliary data of a measure dx on Ω. We require as before that each operator be positive semi-definite and that the collection jointly be a decomposition of the identity

$$Q_x \geq 0 \ \forall x \in \Omega, \text{ and } \int_\Omega dx Q_x = \mathbf{1}. \tag{18}$$

For any $\psi \in \mathbb{C}^d$ and $x \in \Omega$, the probability density of outcomes

$$p_\psi(x) = \langle\psi| Q_x |\psi\rangle \tag{19}$$

gives rise to finite probabilities

$$\text{Prob}_\psi(\text{outcome} \in S) = \int_S dx \langle\psi| Q_x |\psi\rangle \text{ for } S \subset \Omega \text{ measurable} \tag{20}$$

and expected values

$$\mathrm{E}_\psi[f : \Omega \to \mathbb{C}] = \int_\Omega dx \, \langle\psi| \, Q_x \, |\psi\rangle \, f(x) \tag{21}$$

So that all of the above makes analytic sense, the mapping $x \mapsto Q_x$ must be measurable. Notice then that the mapping Q which takes measurable subsets $X \subset \Omega$ to operators

$$Q(X) := \int_X dx Q_x \tag{22}$$

can itself be thought of as a measure on Ω taking positive(-semidefinite)-operator values, justifying the term POVM.

For the particular problem of pure state estimation, even before we decide how to map states $\left|\hat{\phi}\right\rangle$ to operators $Q_{\hat{\phi}}$, we must decide on an appropriate measure. Conjugation by any fixed unitary as a symmetry of all observable physics, so any physically sensible measure must respect this symmetry.

$$\int d\psi f(\psi) = \int d\psi f(U\psi U^\dagger). \tag{23}$$

Just as there is a unique rotation-invariant measure on the sphere up to overall scaling, this symmetry uniquely fixed $d\psi$ to be the *Haar measure* after we fix scaling by demanding that $d\psi$ be a probability measure.